# 📊 Retail Sales Performance & Profitability Analysis Report

## 1. Introduction

The aim of this project is to analyze retail sales data to understand business performance, compare product categories, and study monthly sales trends. This analysis helps in making better decisions related to inventory management, marketing, and profitability.

The tools used in this project are:

- Python (Google Colab) for data cleaning
- SQL for data analysis
- Tableau Public for data visualization and dashboard creation

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## 2. Dataset Description

The dataset contains retail transaction data with the following main fields: Transaction ID, Date, Customer ID, Gender, Age, Product Category (Beauty, Clothing, Electronics), Price per Unit, Quantity, and Total Amount.
Before analysis, the dataset was cleaned to remove missing and incorrect values.

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## 3. Data Cleaning & Preparation

Data cleaning was done using Python (Pandas) in Google Colab. The following steps were performed:

- Removed missing and null values
- Checked and corrected data types
- Created a cleaned dataset file (retail_sales_cleaned.csv)
- Used the cleaned data in SQL and Tableau

This ensured that the data used for analysis is accurate and reliable.

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## 4. SQL Analysis

After cleaning, the data was analyzed using SQL. The main queries used were:

**Total Sales by Category**

```
SELECT Product_Category, SUM(Total_Amount) AS Total_Sales
FROM retail_sales_cleaned
GROUP BY Product_Category;
```

**Average Order Value by Category**

```
SELECT Product_Category, AVG(Total_Amount) AS Avg_Order_Value
FROM retail_sales_cleaned
GROUP BY Product_Category;
```

**Units Sold by Category**

```
SELECT Product_Category, SUM(Quantity) AS Total_Units_Sold
FROM retail_sales_cleaned
GROUP BY Product_Category;
```

**Monthly Sales Trend**

```
SELECT strftime('%Y-%m', Date) AS Month, SUM(Total_Amount) AS Monthly_Sales
FROM retail_sales_cleaned
GROUP BY strftime('%Y-%m', Date)
ORDER BY Month;
```

These queries helped in understanding category-wise performance and monthly sales patterns.

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# 5. Dashboard & Visual Analysis

A Tableau dashboard was created with the following charts:

- Average Order Value by Category
- Units Sold by Category
- Total Sales by Product Category
- Monthly Sales Trend

**Observations:**

- Electronics shows strong total sales and high average order value.
- Clothing sells the highest quantity but has a lower average order value.
- Beauty performs well in terms of average order value.
- Monthly sales fluctuate, showing seasonal trends in customer demand.

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# 6. Key Insights & Recommendations

**Insights:**

- Clothing has the highest sales volume, but Electronics generates higher value per order.
- Sales change month by month, showing seasonality.
- Different categories contribute differently to revenue and quantity sold.

**Recommendations:**

- Focus more on Electronics and Beauty as they generate higher value per order.
- Improve inventory planning for Clothing due to high sales volume.
- Use seasonal trends to plan promotions during low-sales months.

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# 7. Conclusion

This project analyzed retail sales data using Python, SQL, and Tableau. The dashboard provides clear insights into product performance and sales trends. These insights can help businesses improve inventory planning, marketing strategies, and overall profitability.

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# 8. Dashboard Link

https://public.tableau.com/app/profile/zainab.mani/viz/Retail_Sales_Analysis_17716711188820/Dashboard2?publish=yes